

Subtract Probes

☐ !SHAPE	
... blend_maxs	{ 3, 3, 3 }
... blend_mins	{ 3, 3, 3 }
... box	☐ False
... size_max	{ 72, 72, 72 }
... size_min	{ 72, 72, 72 }
... subtract	☒ True

When the "subtract" KVP on reflection probes is set on a child probe, the shape of the child will be subtracted from the area that the parent effects. This overrides and cancels out all other multiboxes that it overlaps. The subtract KVP will not function if it is set on the parent probe. Subtract shapes are rendered in red in both viewports. In order to free up the red color, the incomplete funky probe shapes are now rendered in yellow.

As shown below, subtract probes can be either boxes or funky probes. Unlike regular funky probes, subtract probes do not have to be a closed shape, and can be open-ended and infinite. However you pay for the first six planes, performance-wise, all in one go. So if you add one or two planes, may as well add more if you can use them.

If a subtract probe is a funky probe, the alt-click to set the edges assumes you are clicking the outside of a shape you want to exclude, rather than the inside of a room, so it will orient the plane to face the object you clicked on. Blends will also blend inward instead of outward. The 2D click and drag operates similarly, assuming that the probe center sits outside the shape being drawn so that it will be out in open space instead of buried in whatever area you're trying to exclude. If this is an obstacle for some reason, it is always possible to switch subtract off temporarily to build the shape inside-out, and then switch back. Box-shaped subtract probes are not inverted in any way, because working with them using auto snap would be confusing, however grid-drag and snap in the 2D window still works.

